COMPUTER ENGINEERING · UNIVERSITY OF GEORGIA · EXPECTED MAY 2028

I like building things.

I'm a computer engineering student and undergraduate researcher at the University of Georgia. My work spans embedded hardware, robotics, and systems software, from a motor controller PCB and firmware to Windows driver and backend development.

CURRENT FOCUS

- SERS lab research
- Embedded systems
- Systems programming
- Robotics & autonomy

Selected Projects

Embedded hardware, backend systems, robotics, and computer vision.

EMBEDDED SYSTEMS · MOTOR CONTROL

01

Brushless Motor Controller

C · STM32 · KiCad · SPI

Designed a three-phase motor controller PCB in KiCad and wrote C firmware for PWM, SVPWM, and field-oriented control. The board has USB-C connectivity and supports an optional high-resolution SPI absolute encoder.

Verified PWM timing and deadtime with an oscilloscope, then debugged gate-drive and motor-control behavior on the assembled board.

Firmware | PCB Design | FOC | SVPWM | Hardware Debugging

View project →

SYSTEMS SOFTWARE · SECURITY

02

Soltix.cc

Founder · Four-year independent venture

Founded a commercial cheat-detection service that analyzed Windows process memory and system artifacts. It reached more than 300 users and 37 customers.

Built a Windows driver in C/C++ and process-analysis tools in Go, and administered Linux systems for customer deployments.

Windows Driver | Process Memory | C/C++ | Go | Linux

View project →

ROBOTICS · AUTONOMY

03

FRC Team 1683

Programming Lead · 2 Years · Team Member · 4 Years

Led programming for two of my four years on the team, developing autonomous and computer-vision software for competition robots. The 85-student team reached the FIRST World Championship in both years I served as lead.

Also contributed to robot design, electrical systems, fabrication, and CNC machining.

Robot Software | Autonomous Systems | Computer Vision | Electrical Systems | CNC

View project →

BACKEND DEVELOPMENT · HACKGT 13 04

SideQuests

MongoDB Prize · 1st Place, NSA HEARSAY

Built the Go backend and server infrastructure for a MongoDB-backed app that combines time- and location-aware activity plans with shared outings, interest matching, and group chat.

Go | MongoDB | Backend Development | Server Infrastructure

View project → **Devpost ↗**

COMPUTER VISION · MECHANISM DESIGN 05

Sortify

Google Gemini Track Winner · MakeMIT x Harvard

Developed computer-vision software to classify waste and pass the results to sorting controls. Designed a compact differential mechanism to route each item to the correct bin.

Computer Vision | Sorting Controls | Mechanism Design

View project → **Devpost ↗**

BACKEND DEVELOPMENT · DOCUMENT SEARCH 06

DocDoctor

MongoDB Track Winner · Google Cloud Honorable Mention · AI ATL

Built a Go backend and WebSocket pipeline for live support-call transcripts and agent annotations. Added MongoDB persistence and document-search APIs so agents could retrieve relevant material during calls.

Go | WebSockets | MongoDB | Document Search | APIs

View project → **Devpost ↗**

AI · EDUCATION 07

Sophi

NexHacks · Carnegie Mellon

Built an AI learning tool that generates instructor-style questions from course material and gives targeted hints to help students practice problem solving with material tailored to their class.

AI | Learning Tools | Education

View project → **Devpost ↗**

ROBOTICS · COMPUTER VISION 08

Distributed Planetary Analysis Platform

3rd Place · Autonomous Track · RoboTech

Developed software for a humanoid robot and autonomous vehicle using inverse kinematics, fiducial detection, pathfinding, PWM motor/servo control, and obstacle avoidance. The autonomous vehicle ran on a Jetson and used fiducial tracking to navigate its environment.

Jetson | Computer Vision | Inverse Kinematics | Pathfinding

View project → **Devpost ↗**

RESEARCH & WORK

Experience

B.S. in Computer Engineering at the University of Georgia, expected May 2028. GPA: 3.9/4.0.

01	**UGA SERS Lab**	**Undergraduate Researcher · Sep 2026–Present.** Designed the scanner architecture and modeled the complete mechanical system in Onshape for an automated Raman/SERS platform. Selected and ordered electronic components, corrected prior wiring faults, and contributed to fabrication and full-stack scanner software.
02	**Soltix.cc**	**Founder · Four-year independent venture.** Built commercial Windows process-analysis software, including a C/C++ driver and Go tools, and managed Linux deployments for 37 customers.
03	**FRC Team 1683**	**Programming Lead · Two years.** Led autonomous and computer-vision development on an 85-student team that reached the FIRST World Championship in both years I served as lead.

Technical Skills

Tools and disciplines used in my research, projects, and engineering work.

Languages

C, C++, Go, Python, Java, JavaScript

Engineering

Embedded systems, firmware development, PCB design, electronics, robotics, CAD/CAM, Onshape

Software & systems

Systems programming, Windows driver development, computer vision, backend development, WebSockets, MongoDB, Linux

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